Nexus Widgets Deployment Guide

1. Introduction

Nexus Widgets are small battery-powered sensors that measure temperature, humidity and vibration in warehouses, kitchens and workshops. Each widget reports readings every thirty seconds over a low-power radio link to a hub, and the hub forwards the data to the Nexus cloud. This guide explains how to plan a deployment, install and calibrate the hardware, and read the dashboards.

2. Planning your deployment

Before ordering hardware, map the spaces you want to monitor. Walk each room and note where the most reliable wall or ceiling mount would sit, how far it is from the nearest hub, and whether metal shelving will block the signal. A widget placed behind a steel rack typically needs a second hub, or a repeater, to keep a steady link. Record the floor plan with rough distances, because the planning tool in the dashboard asks for them later.

3. Installing hardware

3.1 Mounting widgets

Mount each widget with the adhesive pad or the screw bracket included in the box. The adhesive pad works well on smooth painted surfaces, but it should not be used on textured plaster or on surfaces that will see temperatures above fifty degrees Celsius. The screw bracket is the better choice for those places. Keep the sensor face unobstructed so that the air around it moves freely, because a widget boxed in behind a panel reports temperatures that run several degrees high.

3.2 Setting up the hub

Power the hub from the included adapter and connect it to the building network with the Ethernet cable. The status light on the hub turns solid green once it has an address and a link to the cloud. A slow blue blink means the hub is waiting for the first widget to join. Pair widgets one at a time by holding the pairing button for three seconds until the widget light flashes white, then bring the widget within one metre of the hub. The pairing completes within about ten seconds, and the widget appears in the device list with its serial number.

4. Calibration

Calibration corrects small offsets that come from manufacturing tolerances and from the mounting position. Place a reference thermometer beside a widget and let both settle for fifteen minutes. In the dashboard, open the device page, choose Calibrate, and enter the reference reading. The widget stores the offset and applies it to every later reading. Humidity calibration uses a saturated salt solution in the same way, and the procedure takes about an hour because the sensor needs time to reach equilibrium. Calibrate every widget once after installation and again after any change of mounting position.

After calibration, check the result on a second reference point. Warm the widget by holding it in a closed hand for two minutes, then compare the reading with a thermometer in a bag beside it. The difference should be under half a degree. If it is larger, repeat the procedure with a fresh reference reading, because a reference that was still warming up is the most common cause of a poor offset. Note the date, the reference used and the final offset in the device logbook so that the history of each widget can be reviewed.

Sensors drift slowly over several years. A widget that was accurate when installed may read half a degree high after three years, which is inside the stated tolerance but can still matter for cold chain records. Schedule a check every twelve months for devices that hold regulated stock, and recalibrate any widget that falls outside the tolerance printed on its label. Devices that are replaced should be calibrated before they join the hub, so that the new serial number starts with a valid offset.

5. Alerts

5.1 Writing rules

Alerts are rules that compare a reading with a threshold for a set duration. A rule named cold storage high, for example, can fire when the temperature stays above four degrees for ten minutes. Rules can be scoped to a single widget, to a group of widgets, or to a whole site. When a rule fires, the dashboard shows the alert in the feed, and the configured channels receive a notification. Each channel can be an email address, a chat webhook or a text message, and every rule can use several channels at once.

5.2 Choosing durations

Avoid very short durations. A duration of thirty seconds will fire on a door opening and closing, which produces noise without helping anyone. Most teams find that five to fifteen minutes gives a good balance between early warning and quiet nights. Snoozing an alert silences its notifications for a chosen period but keeps the event in the feed, so the history stays complete for later review and for audit purposes.

6. Network and security

The hub needs an address from the building network, either by DHCP or a fixed address. It connects outward to the Nexus cloud over HTTPS on port 443, and it does not need any inbound port to be opened. If your firewall inspects traffic, allow the hostname printed on the hub label and disable content rewriting for that host. The hub checks the time against a network time server on start-up, so an outbound route for NTP is also useful. Readings are buffered on the hub for up to six hours if the cloud link drops, and they are sent in order once the link returns.

7. Maintenance

Routine maintenance takes less than an hour per site. Every quarter, wipe the sensor faces with a dry cloth to remove dust, which can slow the response of the humidity sensor. Check the battery indicator on each widget and replace any battery that has fallen below twenty percent, using only the cell type listed on the label. Apply the latest firmware from the dashboard, and confirm on the hub page that every widget reports the same firmware version. Keep a short written record of the visit so that the maintenance history is visible to the whole team.

8. Data retention

Readings are kept for ninety days at full resolution and for two years as hourly averages. Alert events are kept for one year. Administrators can export any of these ranges as CSV or as a JSON file, and exports are available for thirty days after they are created. Deleting a device removes its readings from the dashboard immediately, and the deletion is completed in the backups within thirty days. Account owners can request a full deletion of an account, and that request is confirmed by email before it is carried out.

9. Roles and access

Access is controlled by three roles. Viewers can see devices, readings and alerts but cannot change anything. Operators can add and remove widgets, acknowledge alerts and run calibrations. Administrators also manage members, alert channels and the account plan. Assign the smallest role that a person needs, and review the member list every quarter, because people change teams and their access should change with them.

10. Glossary

Widget. A battery-powered sensor that measures temperature and humidity and, on some models, vibration. Each widget has a serial number printed on its label and a unique identifier in the dashboard. Widgets never talk to the cloud directly; they always report through a hub.

Hub. The gateway that joins widgets to the building network and forwards their readings to the cloud. A hub buffers readings during outages, keeps the widget list, and applies firmware updates. Each site needs at least one hub, and large sites may need more than one.

Calibration offset. A correction value stored on a widget and applied to every reading it produces. Offsets are measured against a reference instrument and remain in place until the widget is calibrated again or removed from the device list.

11. Frequently asked questions

Can widgets share a hub with other radio equipment? Yes, as long as the other equipment does not use the same channel for long periods. The hub chooses a quiet channel at start-up and moves to another one if interference is detected, which takes about a minute and does not lose any buffered readings.

What happens when the battery of a widget is replaced? The widget keeps its serial number, its calibration offset and its alert rules. Readings resume within one reporting interval once the new battery is seated and the status light has flashed white twice.

Is there a way to test an alert without waiting for a real event? Yes. Each rule has a Test button that sends a sample notification through the channels attached to that rule. The test notification is marked clearly as a test and does not appear in the event history.

12. API and troubleshooting

The Nexus API exposes devices, readings and alerts over HTTPS with JSON bodies. Requests are authenticated with a bearer token that you create in the account settings. Readings are paginated by timestamp, and the largest page size is one thousand rows. Rate limits are generous for dashboards and integrations, but bulk exports should use the export endpoint, which returns a file that can be downloaded once it is ready.

If a widget stops reporting, first check the battery indicator on the device page. Below ten percent the widget reduces its transmission rate to conserve power, which looks like a gap in the chart. Next, check the distance to the hub and whether a new metal object has been placed between them. A widget that has been powered off and on again rejoins the hub automatically, and the gap closes within one reporting interval.